Methods Supplement: Identity, Classification, and Selection Provenance

From Hairballs to Hypotheses

Working supplement, 11 October 2026

This note accompanies the shortened manuscript. It preserves classification and identity detail and incorporates the repository selection audit supplied on 9 October 2026. It has not yet been incorporated into a new Zenodo deposit.

1 Deduplication and Related Reports

Deduplication consolidated 187,446 source occurrences into 140,959 bibliographic records. A *source occurrence* is an entry returned by a provider or imported from another source; several occurrences may refer to the same publication. A *canonical record* is the representative bibliographic entry retained after applying the identity rules.

Matching used normalized DOI keys where available and exact normalized-title keys otherwise. Records with and without DOIs were kept separate when title agreement alone did not establish their identity. arXiv identity adjudications were incorporated into the final total. Source occurrences and identity decisions were retained so that canonical records could be traced to their acquisition routes.

Fifty related-version links connect publications such as preprints and published articles. These links preserve relationships between reports that remain separate bibliographic records. Grouping reports about the same software system occurs during full-text assessment, as described in the main paper.

2 Provisional Candidate Classification

The task groups, assistance modes, and visualization modalities are defined in Section 3 of the main paper. Each label was assigned a Present, Absent, or Uncertain state, and multiple labels could coexist. Classification used evidence for the specific category: for example, unspecified display information yielded uncertainty about modality. The same category definitions were applied to all 9,505 advanced records using their supplied titles and abstracts.

These provisional classifications organized the candidate pool and guided selection for full-text assessment. Table 1 summarizes assistance–modality co-occurrences among candidate records. A record contributes to a cell when both labels are Present; cells overlap. In the export, Present and Uncertain labels are listed explicitly, with omission denoting Absent under the classification contract. Desktop/Planar, for example, was Present in 2,912 records and Uncertain in 4,072.

The table describes labels supported by bibliographic metadata. Establishing that a particular assistance mechanism operates within a particular display environment requires assessment of the system description, which underlies Figure 3 of the main paper.

Table 1: Machine-coded candidate co-occurrences among 9,505 records. Both labels must be PRESENT; cells overlap and are not additive. These are discovery counts, not verified mechanism placements. D: Desktop/Planar; L: Large Display; A: AR/MR.

Assistance	D	L	VR	A	CAVE
Algorithmic	2,837	37	511	399	11
Adaptive	806	12	185	190	2
Conversational	86	1	15	22	0
Immersive	112	20	428	346	8

3 Purpose-Specific Selection Rules

The 384 selected report identities represent six sets and 381 systems, not 384 completed assessments or one uniform sample. Of the report identities, 376 link exactly to the terminal 9,505-record pool. Three additional systems entered through coauthor-supplied reports and five through bounded early map expansion. A distinct prior-survey-only full-report selection route is not established by the cumulative crosswalk.

3.1 Initial 20 entries

Purposive anchors informed by earlier model outputs and discussion were combined with assessed coauthor reports to establish the mechanism map. The candidate export resolved identities of already chosen examples, rather than ranking them. The lead author supplied the twelve anchor nominations; a Codex agent nominated the replacement and seven additions from coauthor-source evidence. The initial 20 was a resulting set, not a documented numerical cap. The intellectual origin of the supplied anchors and any formal ordering or tie-breaking rule remain unresolved.

3.2 Bounded 18 entries

A frozen queue excluded previously represented systems and prioritized reusable lawful local reports, machine-coded candidates for empty cells, sparse Large Display/AR/MR/CAVE cases with explicit mechanism leads, and bounded contrast cases. Per-entry rationales and queue order are retained. The lead author specified the priorities and a maximum of 40; a Codex agent authored the exact 18-entry queue. Within-priority tie rules and the reason for stopping at 18 remain undocumented.

3.3 Expanded 50 entries

Software ranked candidates after excluding prior identities. Scores reflected rare modalities, non-desktop Adaptive/Conversational labels, mechanism/evaluation terms, contrast terms, and access indicators; ordering used descending score then lowercase title. The final 50 were a separately specified list with per-entry rationales, not the ranked top 50. Software validated and materialized that list within a maximum of 60. A Codex agent curated the final nominations under the lead author's maximum of 60. Rules for choosing among similarly suitable records and the reason for selecting exactly 50 remain undocumented.

3.4 Deterministic 200 entries

Software validated the 9,505-record input and excluded previously considered canonical IDs, exact titles and DOIs, and subsequent exact DOI/title duplicates. Scores combined sparse-cell coverage, workflow, life-science, interaction, computation, evaluation/mechanism terms, abstracts, likely open access, assistance/modality bonuses, and contrast indicators. The implemented

ordering was high-precision signal, rationale class, likely-open-access status, descending score, then canonical identifier. The lead author authorized the 200-entry bound and limit of 140 completed new assessments. The first 200 were frozen: 1–140 were primary and 141–200 ordered reserves. Retrieval continued through queue exhaustion; the maximum of 140 concerned completed new assessments rather than selected report identities. The original audit JSON's textual sort description omits the high-precision and open-access strata. A dated correction overlay, `queue_selection_audit_correction_20261009.json` in the same staging directory, records the implemented order and binds it to the executable script. The original audit remains unchanged; no selection was rerun and no queue membership, order, or assessment changed.

3.5 Sparse-cell 24 entries

An upstream 25-entry shortlist from an 8,832-record partial export allowed up to three candidates per sparse cell, favored abstracts and DOIs, and excluded prior exact IDs/titles. Software rebound the entries to the terminal 9,505 export and checked canonical IDs, DOI, normalized title, and related versions. Removing AREA 2013 as a related report of previously considered AREA 2012 yielded 24 entries in preserved priority order, within a maximum of 30. The lead author supplied the provisional 25-entry list and authorized the maximum of 30; software performed reconciliation. The original intellectual source of the supplied list and within-cell tie rules remain unknown.

3.6 Complete target-cell pass: 72 new entries

Software enumerated every exact Present-labelled record in Immersive × Desktop/Planar (112 memberships), Immersive × Large Display (20), and Adaptive × Large Display (12). The 144 memberships represented 131 records, including 58 prior exact identities. All entered processing; priority band, descending metadata score, and canonical identifier determined access order only. Identity reconciliation reduced them to 129 groups. Excluding already represented systems and related reports yielded 72 new systems in the cumulative selected set. The lead author specified and authorized the three target cells. The 72 new entries were the reconciliation outcome, not an inclusion quota.

4 Access and System-Level Reconciliation

Set	Reports	Systems	Full text	Abstract	Unavailable	Assessed
20	20	20	20	0	0	20
18	18	18	15	0	3	15
50	50	50	25	0	25	25
200	200	197	60	1	139	57
24	24	24	1	0	23	1
72	72	72	14	0	58	14
Total	384	381	135	1	248	132

Two related-report pairs and an iCAVE preprint linked to an already represented system account for the report–system difference. At the September cutoff, the 132 completed assessments comprise 80 eligible, 19 unresolved, and 33 excluded/contextual systems. Unavailable reports remain access outcomes rather than scientific exclusions.

An October access follow-up supplied three previously unavailable reports within the same selected set: VIPER (Algorithmic × Desktop/Planar), NAVRNA (Immersive × Large Display, supported by tangible constrained deformation), and Cosmic View (Algorithmic × Large Display, using a planetarium dome). [3, 1, 2] These author-confirmed assessments bring the totals to 138 accessible report entries, 245 unavailable entries, one abstract-only entry, and 135 completed

system assessments (83 eligible, 19 unresolved, and 33 excluded/contextual). Three additional placements bring Figure 3 to 79 represented systems and 122 placements. The six original selection sets and their membership were not rerun or expanded. The September table above remains the original dated reconciliation.

5 Trace and Authority Limits

The report–system crosswalk at `evidence/selection-flow/unit_crosswalk.csv` records assessment set, queue identifier, route, canonical linkage, report identity and version, access, system outcome, and placement qualifications. Wave-specific scripts, frozen queues, source bindings, retrieval manifests, and assessment ledgers provide supporting trace; dedicated execution logs were not found for the static queues.

Contemporaneous session evidence identifies the requesting user's direction of priorities, bounds, and target cells and distinguishes Codex nominations from deterministic software operations. Morris Chukhman retrospectively confirmed these roles on 9 October 2026. The repository record at `docs/full_report_selection_authority.md`, linked from `docs/reproducibility.md`, separates this retrospective confirmation from contemporaneous evidence and preserves unresolved shortlist origins.

Nomination authority, automated assessment, later author review, and independent validation are distinct. No separate author approval of the exact constructed queues before assessment is documented. The lead author has reported reviewing and confirming the Figure 3 placements; this later review supplies neither dated row-level approval records nor independent double coding or pool-wide screening validation. The crosswalk does not provide a reason for every other candidate's non-selection.

6 Bounded Language-Evidence Follow-up

The initial system assessment used accessible English-language reports. Language was not a provider-query or title-and-abstract screening gate. A subsequent audit distinguished provider-reported language, title/abstract language signals, and full-report language; an English abstract was not treated as proof of an English report. The 9,505-record candidate export contains no language column.

A 147-report lead set was ranked for bounded author review. Seven active Tier A reports were assessed; Trellet's French thesis remains separately pending passage verification by Amira Kefi. Marchianò's thesis, Chemoface, and Campoalegre's thesis have English bodies despite their bibliographic language signals; Le Baher's thesis, REDPath, AIMA, and Wang's VisConnectome report have French, German, Spanish, and Chinese bodies, respectively. Six reports received provisional eligibility recommendations, including a qualified infrastructure-boundary case; VisConnectome's substantive computational-assistance criterion remains unresolved. These are assistant recommendations pending author adjudication, not seven approved systems or additions to the manuscript's counts. Companion/version relationships and consequential translated passages require verification. This purposive follow-up does not estimate screening reliability or establish exhaustive non-English coverage.

7 Task-Symbol Interpretation

Task letters identify navigation (N), selection/filtering (F), comparison (C), sensemaking (S), and coordination/collaborative reasoning (R). Coding follows descriptions of the implemented analytical workflow, not a demonstrated benefit or an interface affordance alone. Coordination can involve task assignment, handoff, or collective interpretation; it need not imply synchronous

multiuser operation. Mechanism–environment placement evidence and task support are distinct. Retained placement rows include analytical-task descriptions, source locators, and qualifications. Repeated task letters do not establish equivalent support, performance, or evaluation across systems.

References

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[3] Philippe St-Jean, Abbas F. Sadikot, Louis Collins, Diego Clonda, Reza Kasrai, Alan C. Evans, and Terry M. Peters. Automated atlas integration and interactive three-dimensional visualization tools for planning and guidance in functional neurosurgery. *IEEE Transactions on Medical Imaging*, 17(5):672–680, 1998.